第5章 模型选择

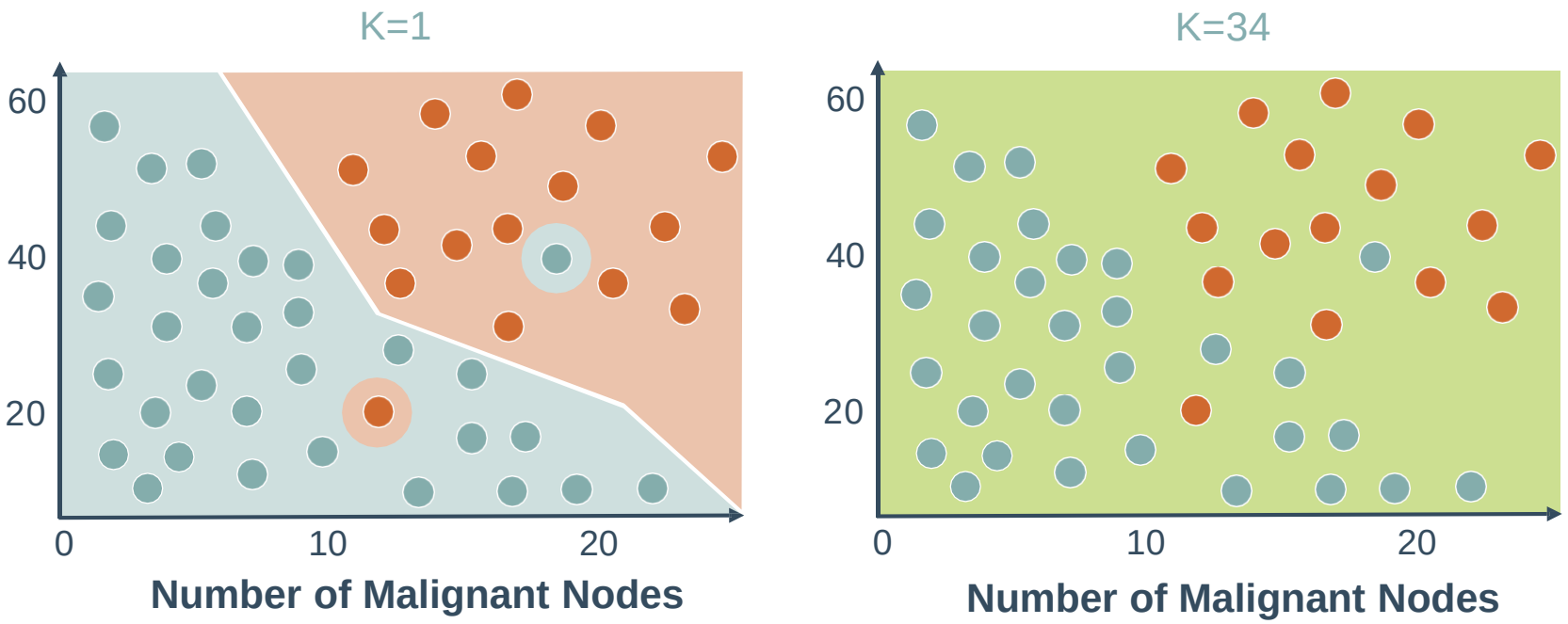
如何选择模型？

对一个给定的有监督学习任务，应该选择哪个学习模型？

如何选择该模型的最优参数？

如何估计训练好的模型在学习样例之外的数据上可能的性能？

K值会影响判定边界


K=1
60
40
20
0
10
20
Number of Malignant Nodes
K=34
60
40
20
0
10
20
Number of Malignant Nodes

训练精度（training accuracy）

- 在整个数据集上训练模型
- 并在同一个数据集上测试模型，得到模型的预测结果，和真实结果做比较，计算模型的精度

问题：

- 机器学习的目标是期望模型能**在学习样例之外的数据上**有好的表现（面向未来，而不是过去）
- 最大化训练精度，通常会产生过于复杂的模型，从而导致过拟合，模型不能很好地**泛化**

泛化误差vs. 经验误差

泛化误差：在“未来”样本上的误差

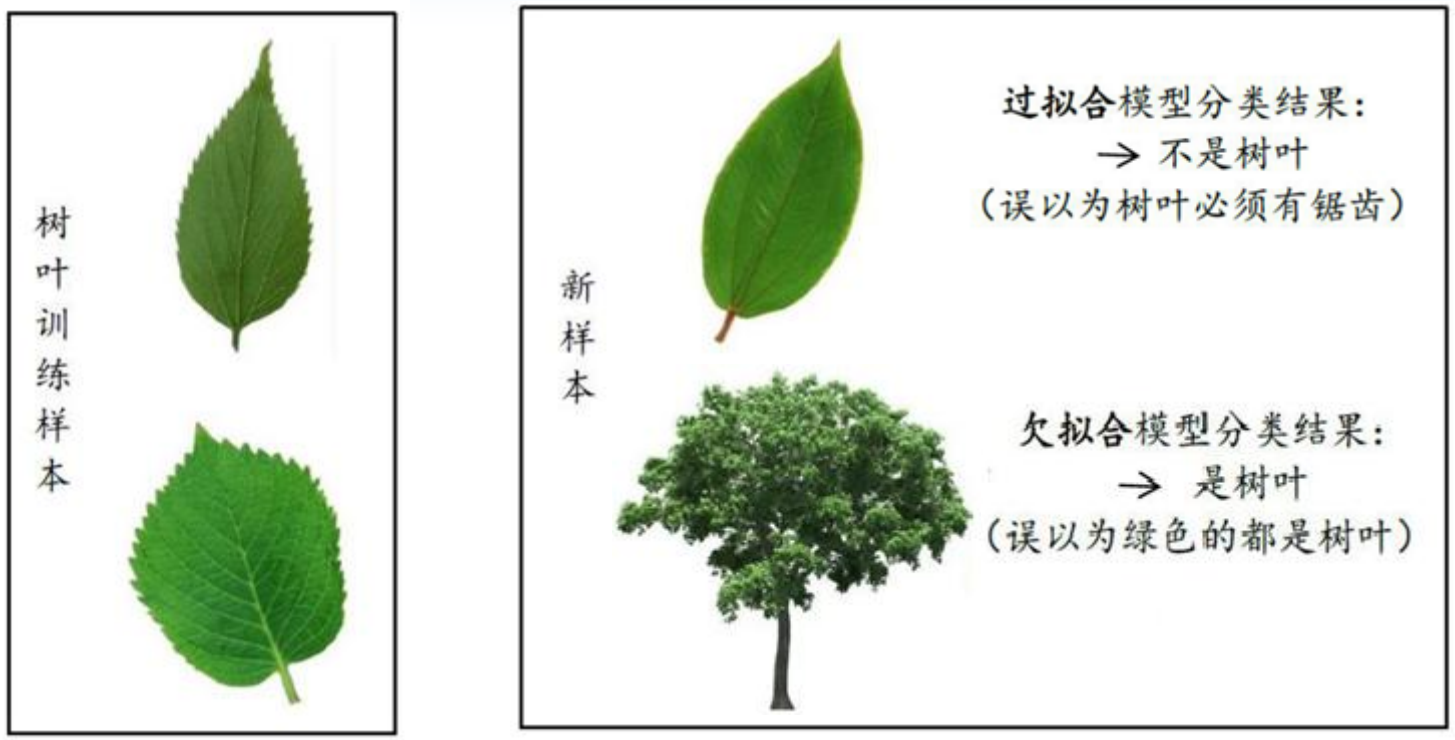
- 泛化误差越小越好

经验误差：在训练集上的误差，亦称“训练误差”

- 经验误差是否越小越好？

NO! 因为会出现“过拟合”(overfitting)

过拟合与欠拟合

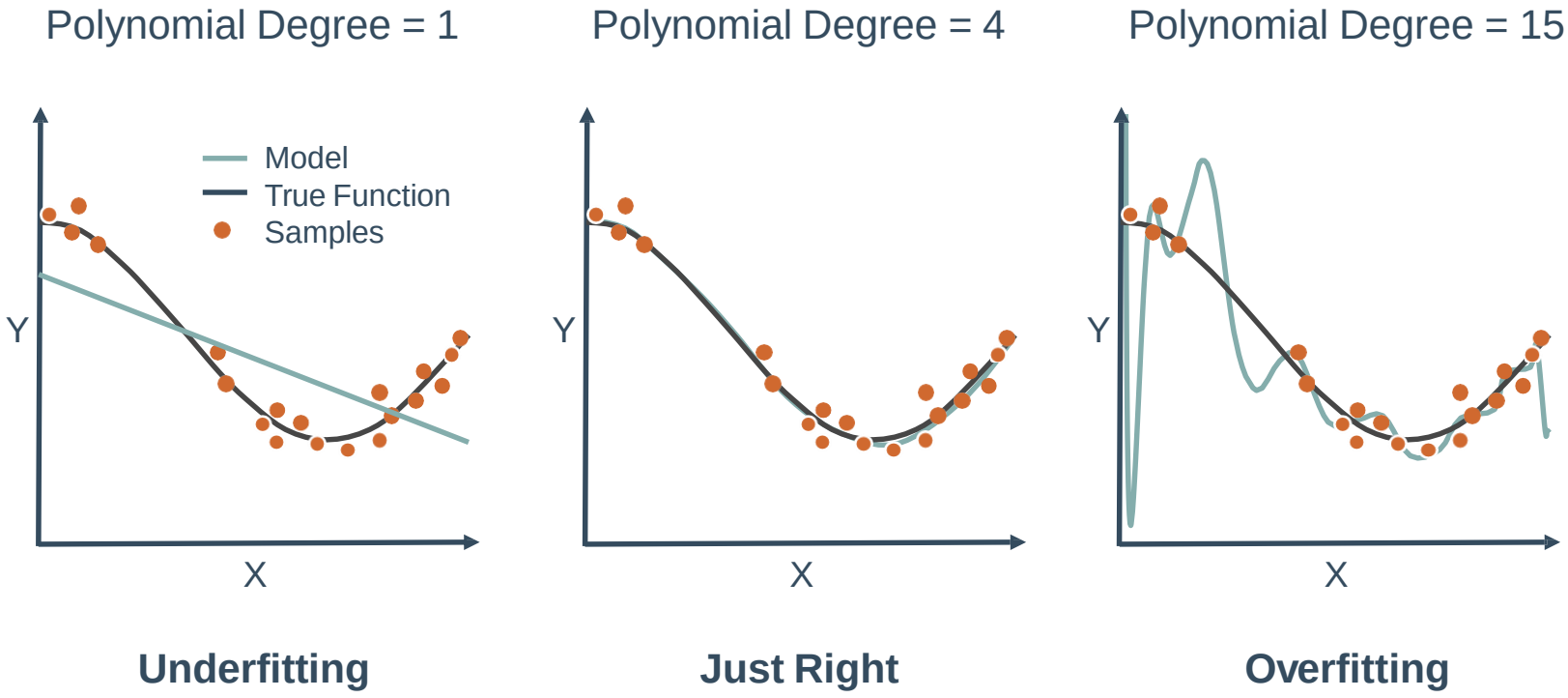


过拟合、欠拟合的直观类比

过拟合：学习器把训练样本本身特点当做所有潜在样本都会具有的一般性质.

欠拟合：训练样本的一般性质尚未被学习器学好.

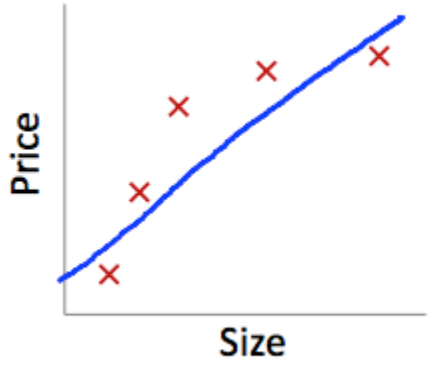
欠拟合与过拟合



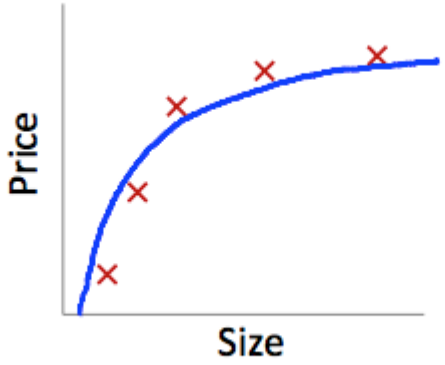
欠拟合和过拟合都会导致较大的泛化误差。

欠拟合一般表示模型对数据的表现能力不足，通常是模型的复杂度不够，并且Bias高，训练集的损失值高，测试集的损失值也高。

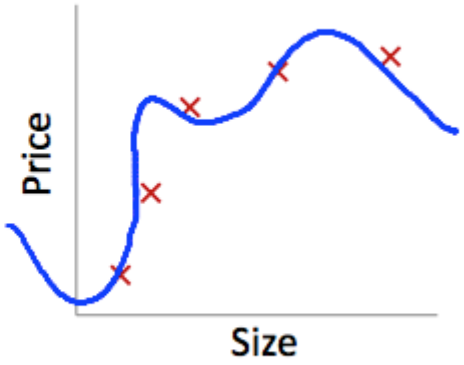
过拟合一般表示模型对数据的表现能力过好，通常是模型的复杂度过高，并且Variance高，训练集的损失值低，测试集的损失值高。



$\theta_0 + \theta_1 x$



$\theta_0 + \theta_1 x + \theta_2 x^2$



$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

偏差-方差分解 (bias-variance decomposition)

对回归任务，泛化误差可通过“偏差-方差分解”拆解为：

$$E(f;D) = \underbrace{bias^2(\boldsymbol{x})}_{} + \underbrace{var(\boldsymbol{x})}_{} + \underbrace{\varepsilon^2}_{}$$

期望输出与真实输出的差别

$$bias^2(\boldsymbol{x}) = \left(\bar{f}(\boldsymbol{x}) - y\right)^2$$

同样大小的训练集的变动，所导致的性能变化

$$var(\boldsymbol{x}) = \mathbb{E}_D\left[\left(f(\boldsymbol{x};D) - \bar{f}(\boldsymbol{x})\right)^2\right]$$

训练样本的标记与真实标记有区别

表达了当前任务上任何学习算法所能达到的期望泛化误差下界

$$\varepsilon^2 = \mathbb{E}_D\left[(y_D - y)^2\right]$$

监督学习中的误差来源

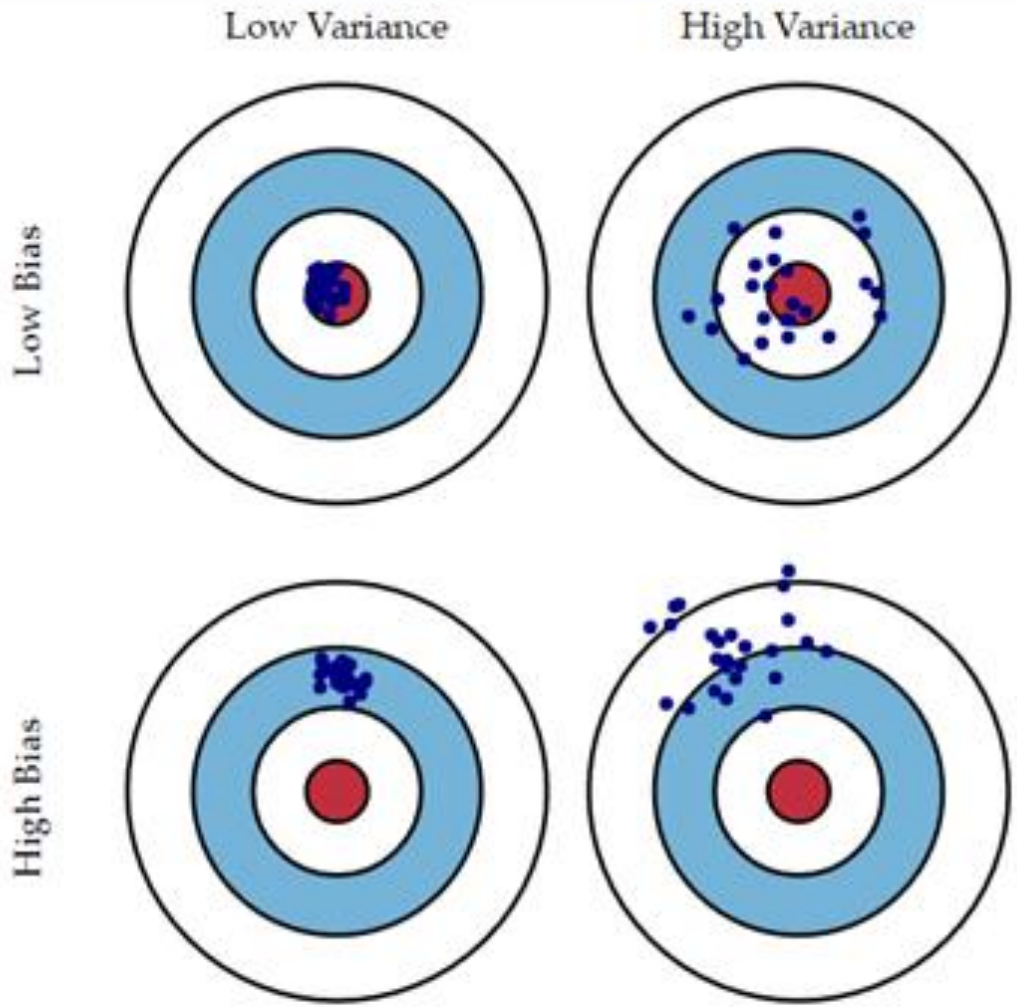
$$Error = Bias^2 + Variance + Noise$$

偏差（Bias）：模型的期望输出值（即用不同数据集训练出的所有模型输出的平均值）与真实值之间的差异。即学习算法的期望预测与真实结果的偏离程度，刻画了学习算法本身的拟合能力。

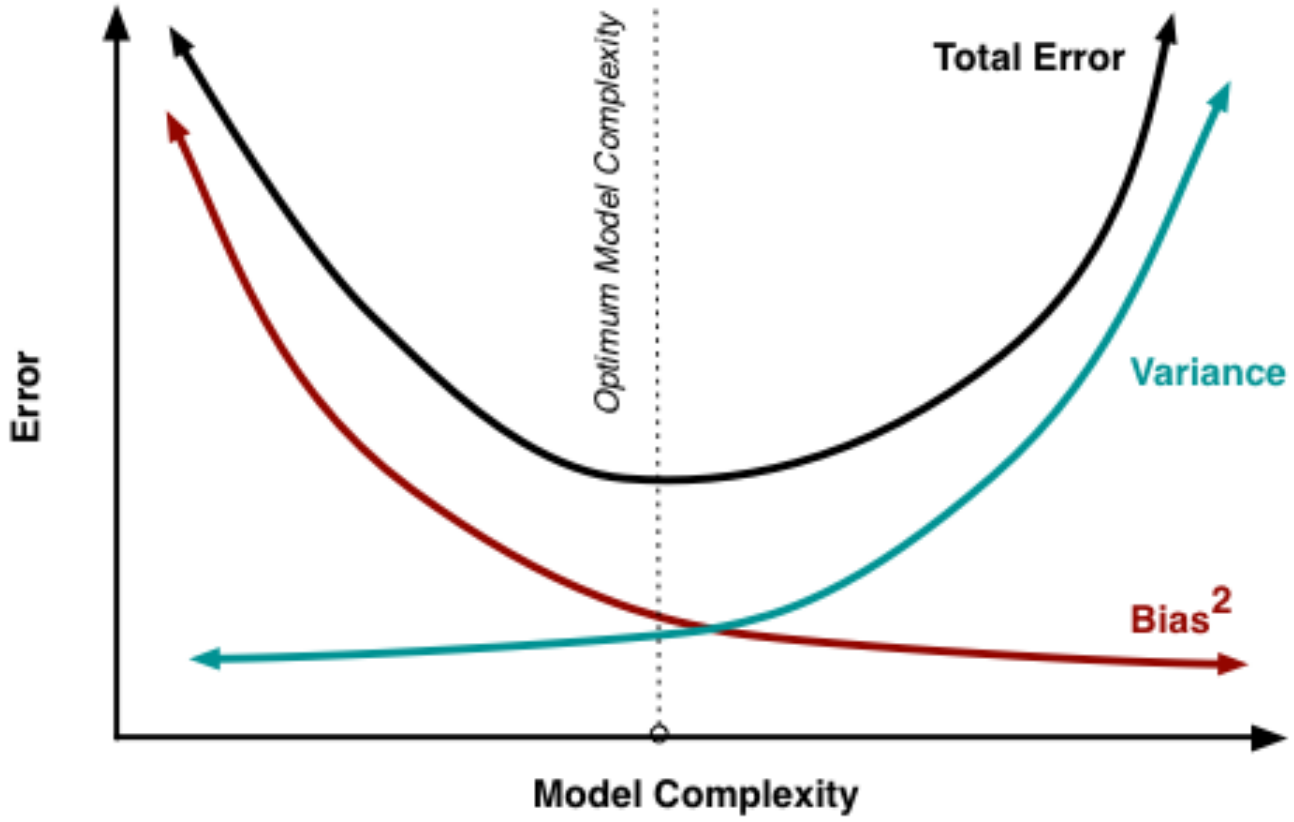
方差（Variance）：用不同数据集训练出的模型的输出值之间的差异。即数据的变动所导致的学习性能的变化，刻画了学习算法的稳定性。

噪声(Noise)：表达了在当前任务上任何学习算法能达到的期望泛化误差的下界，即刻画了学习问题本身的难度。

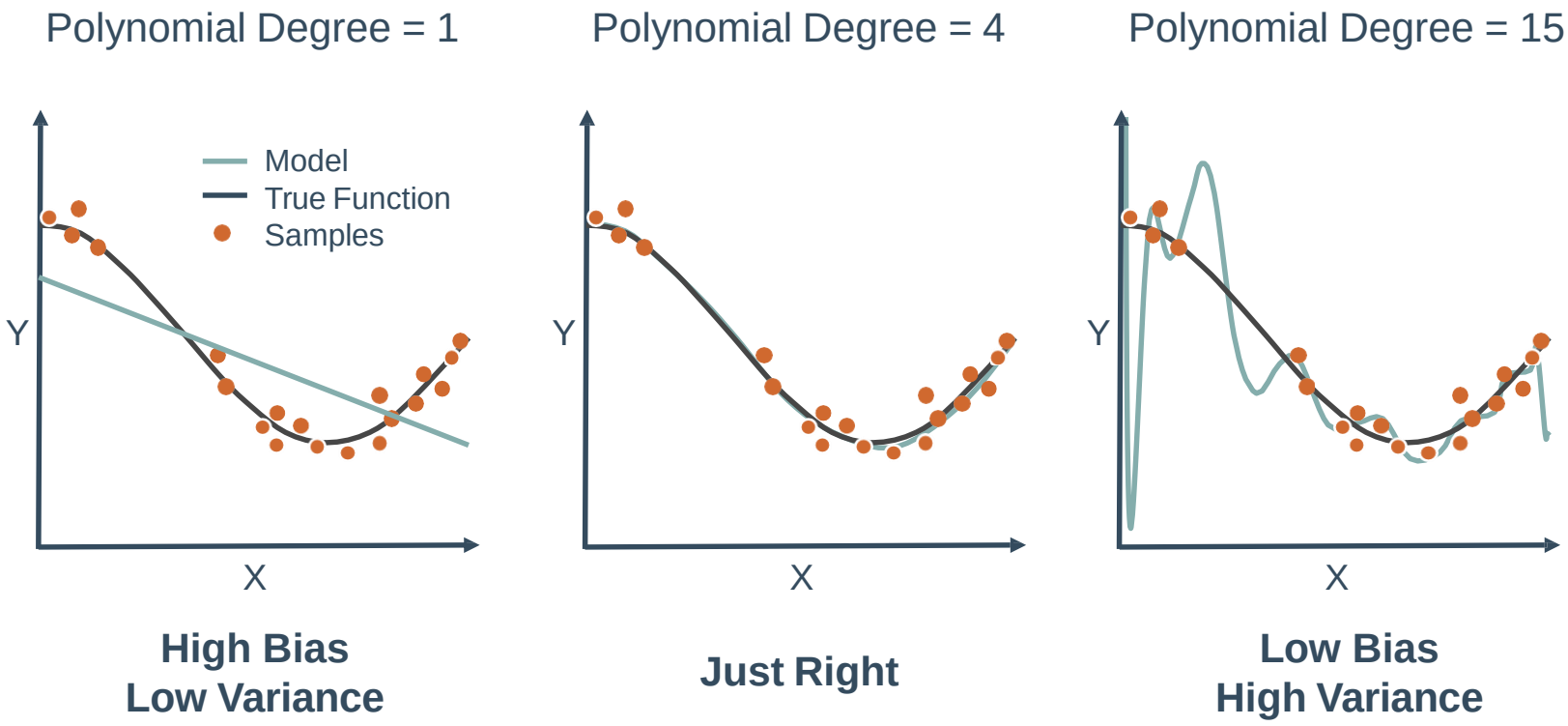
偏差与方差



偏差-方差权衡


Total Error
Optimum Model Complexity
Variance
Bias^2
Error
Model Complexity

偏差-方差权衡


Polynomial Degree = 1
Polynomial Degree = 4
Polynomial Degree = 15
Model
True Function
Samples
Y
X
High Bias
Low Variance
Just Right
Low Bias
High Variance

模型选择

三个关键问题:

- 如何获得测试结果？ ⇨ 评估方法

评估方法

考虑因素

- **泛化性能**
- 时间开销
- 存储开销
- 可解释性

划分训练集和测试集

	Date	Title	Budget	DomesticTotalGross	Director	Rating	Runtime
0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
1	2013-05-03	Iron Man 3	200000000	409013994	Shane Black	PG-13	129
2	2013-11-22	Frozen	150000000	400738009	Chris BuckJennifer Lee	PG	108
3	2013-07-03	Despicable Me 2	76000000	368061265	Pierre CoffinChris Renaud	PG	98
4	2013-06-14	Man of Steel	225000000	291045518	Zack Snyder	PG-13	143
5	2013-10-04	Gravity	100000000	274092705	Alfonso Cuaron	PG-13	91
6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
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10	2013-05-16	Star Trek Into Darkness	190000000	228778661	J.J. Abrams	PG-13	123
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16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

通常将包含个m样本的数据集$D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$
拆分成**训练集** S 和**测试集** T：

将测试集上的“测试误差”作为泛化误差的近似

假设测试集是从样本真实分布中独立采样获得

测试集与训练集尽量互斥

使用训练集和测试集

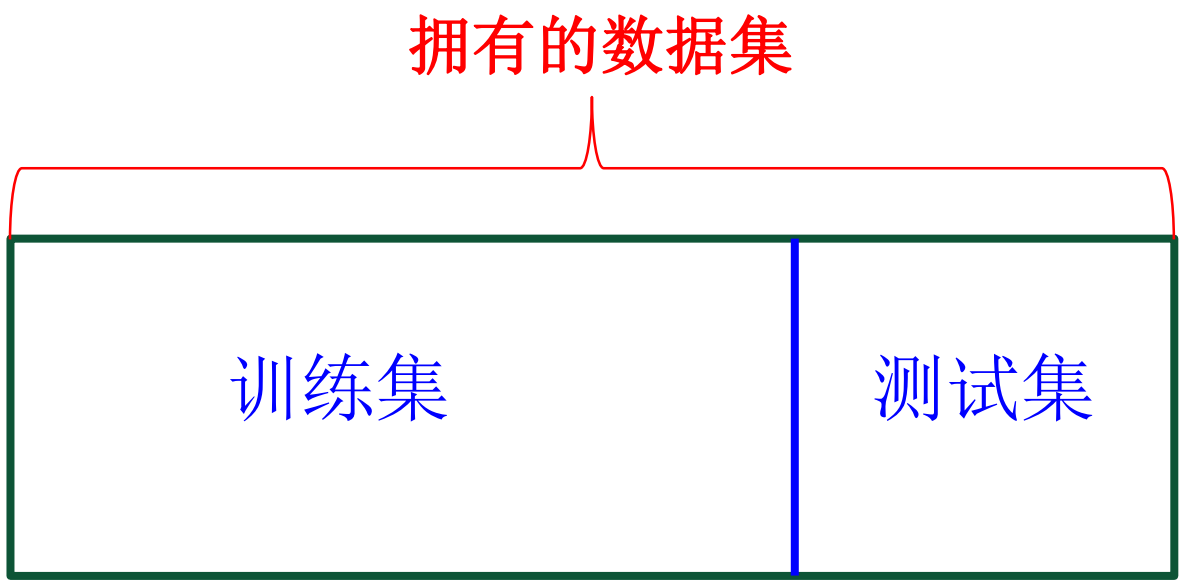
训练数据	训练模型
测试数据	评价模型 - 用模型预测类别标签 - 和真实值比较 - 计算误差

评估方法

- 留出法 (hold-out)
- 交叉验证法 (cross validation)
- 自助法 (bootstrap)

留出法：

- 直接将数据集划分为两个互斥集合



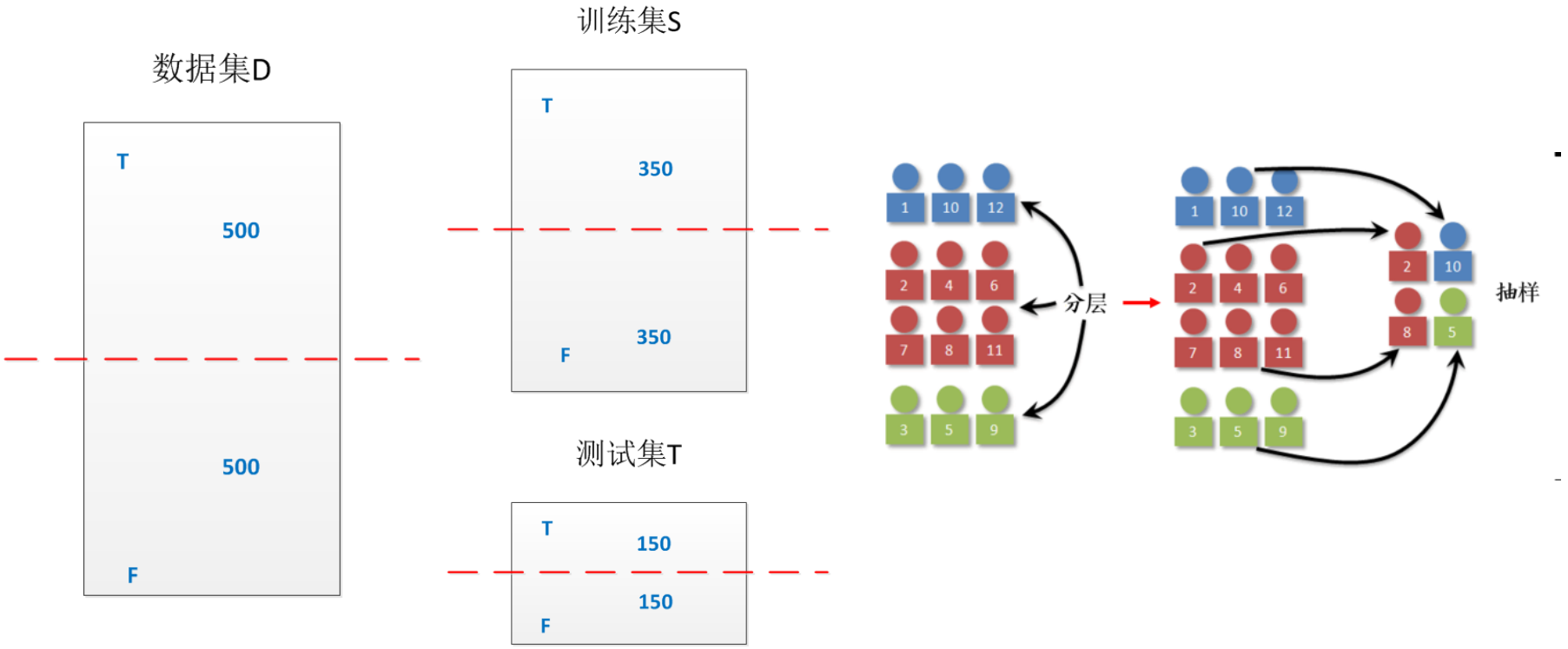
将测试集上的“测试误差”作为泛化误差的近似

假设测试集是从样本真实分布中独立采样获得

尽量保持数据分布一致性

分层采样：保留类别比例

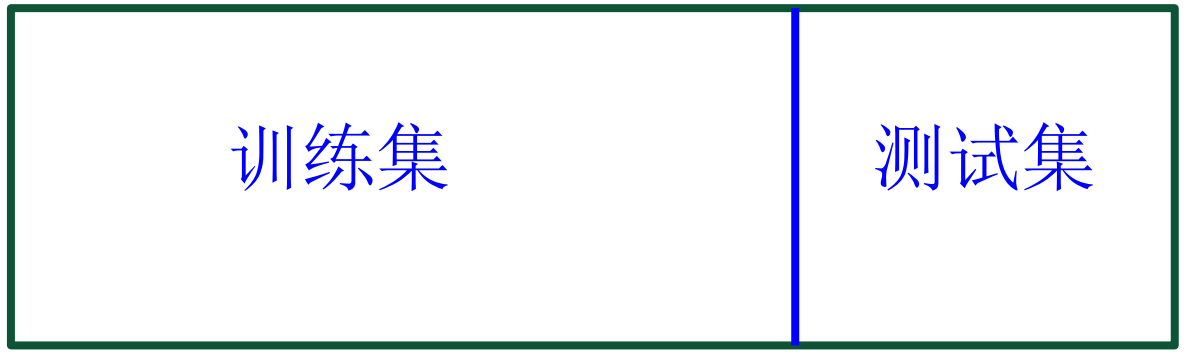
先将总体按某种特征分为若干次级总体（层），然后再从每一层内进行随机抽样，组成一个样本。



一般若干次随机划分、重复实验取平均值

测试集的大小？

训练/测试样本比例通常为3:1~5:1



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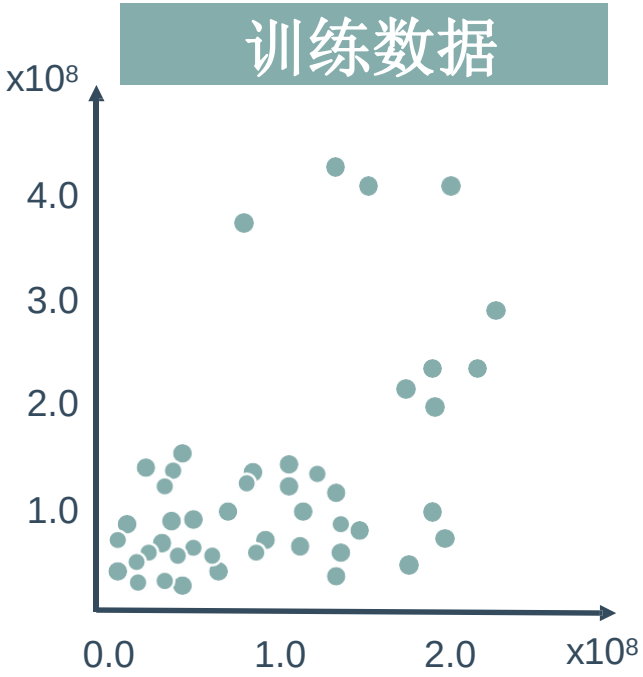
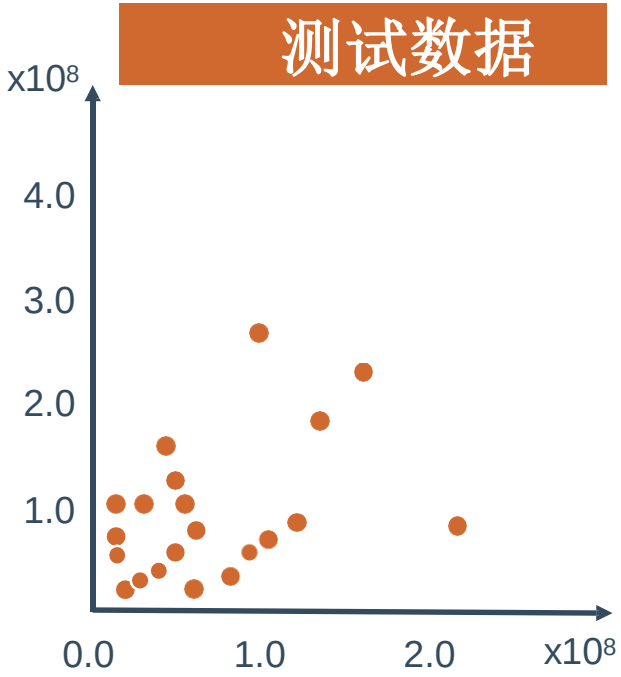
训练数据

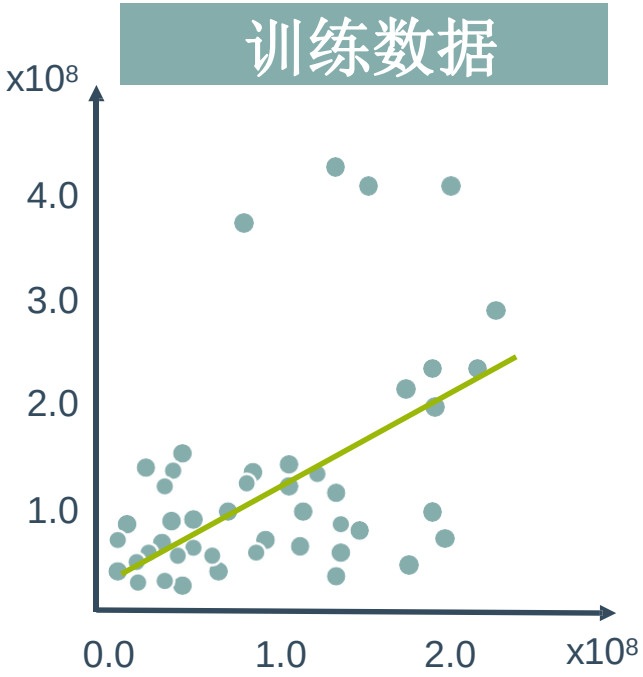
测试数据

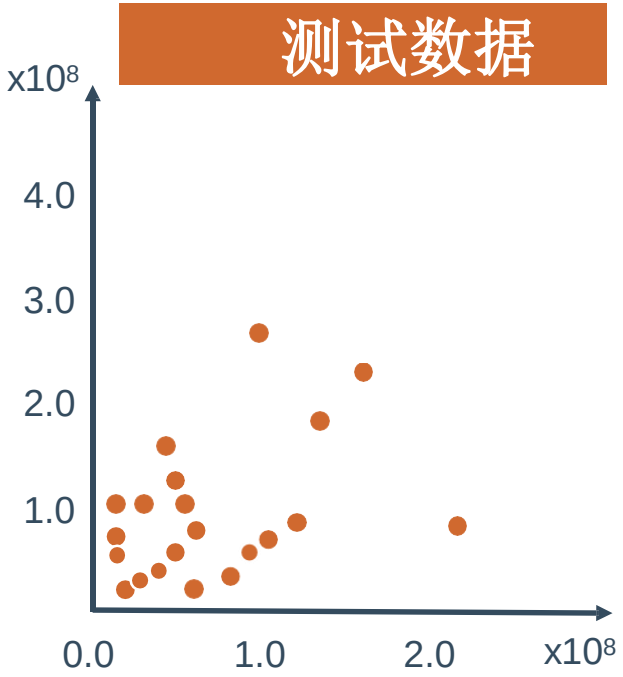
使用训练集和测试集


训练数据
x10⁸
4.0
3.0
2.0
1.0
0.0
1.0
2.0
x10⁸


测试数据
x10⁸
4.0
3.0
2.0
1.0
0.0
1.0
2.0
x10⁸

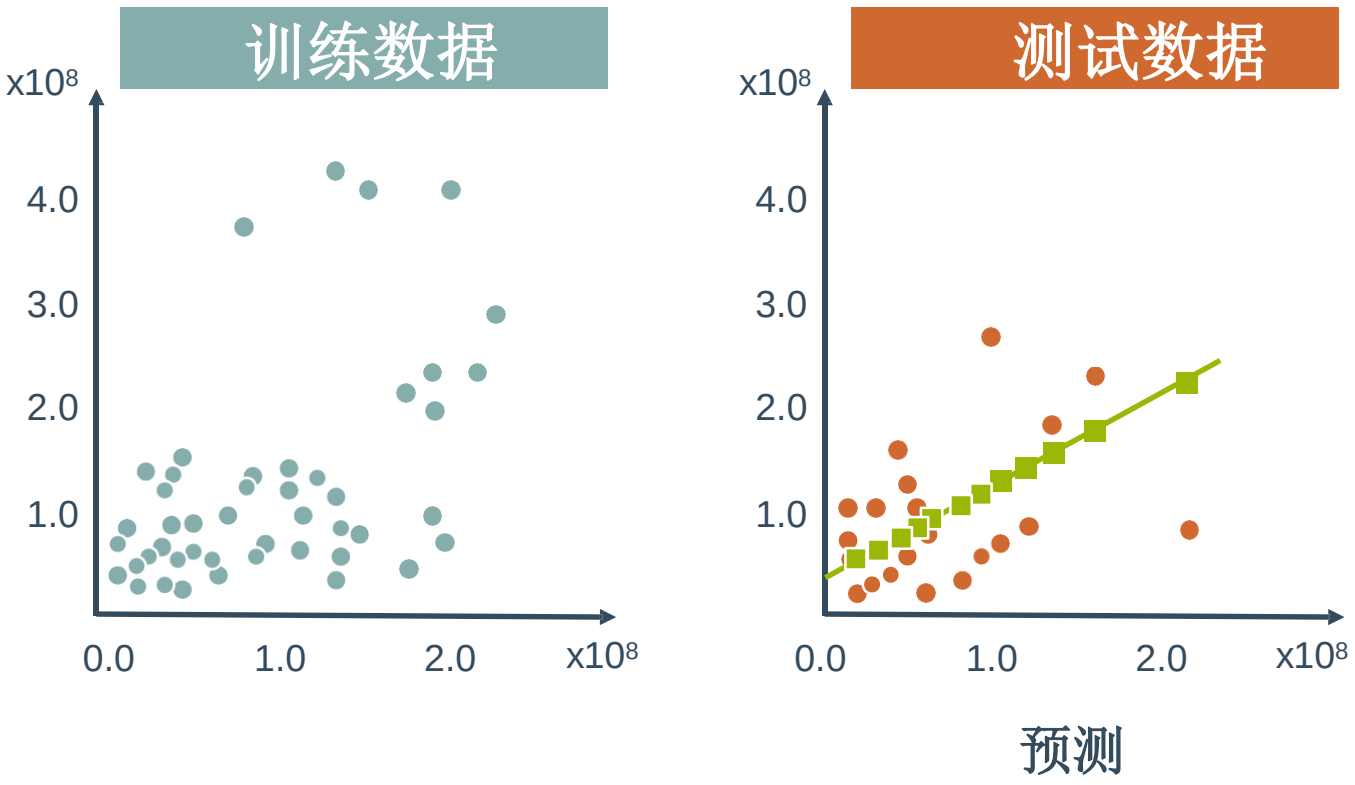
使用训练集和测试集



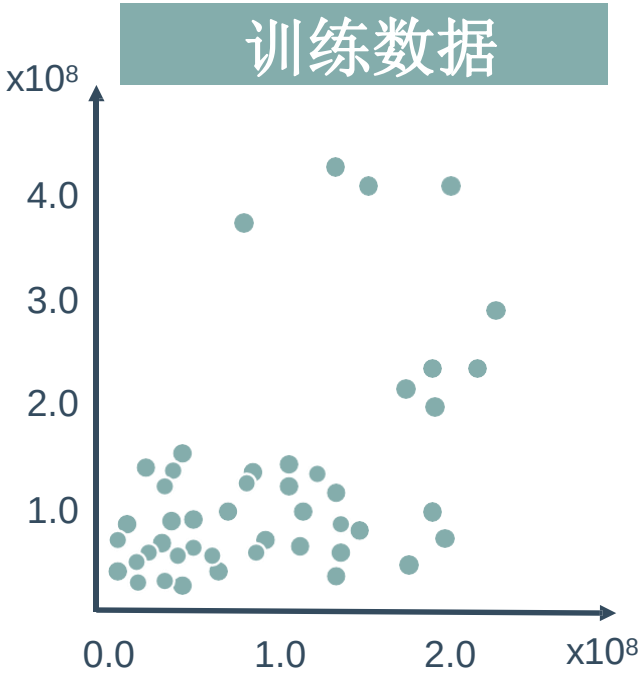


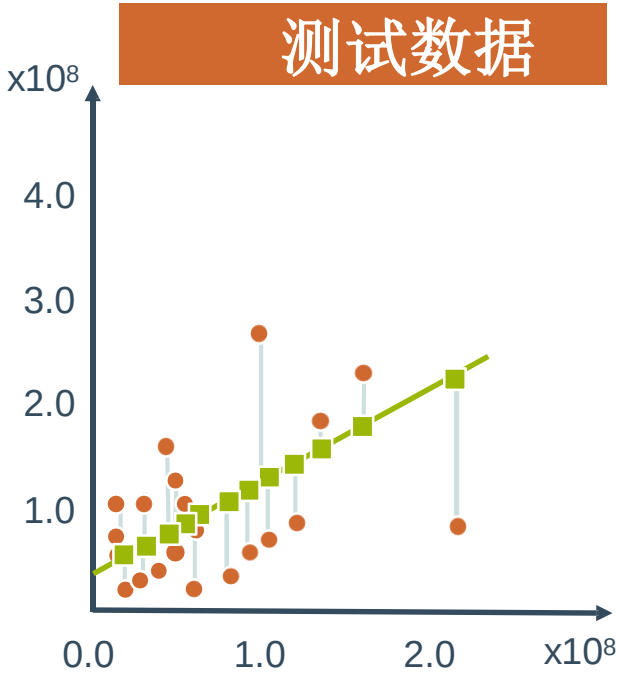
训练模型

使用训练集和测试集


训练数据
x10⁸
4.0
3.0
2.0
1.0
0.0
1.0
2.0
x10⁸
测试数据
x10⁸
4.0
3.0
2.0
1.0
0.0
1.0
2.0
x10⁸
预测

使用训练集和测试集





计算误差
（或精度）

划分训练集和测试集的语法

导入划分训练集和测试集的函数：

```
from sklearn.model_selection import train_test_split
```

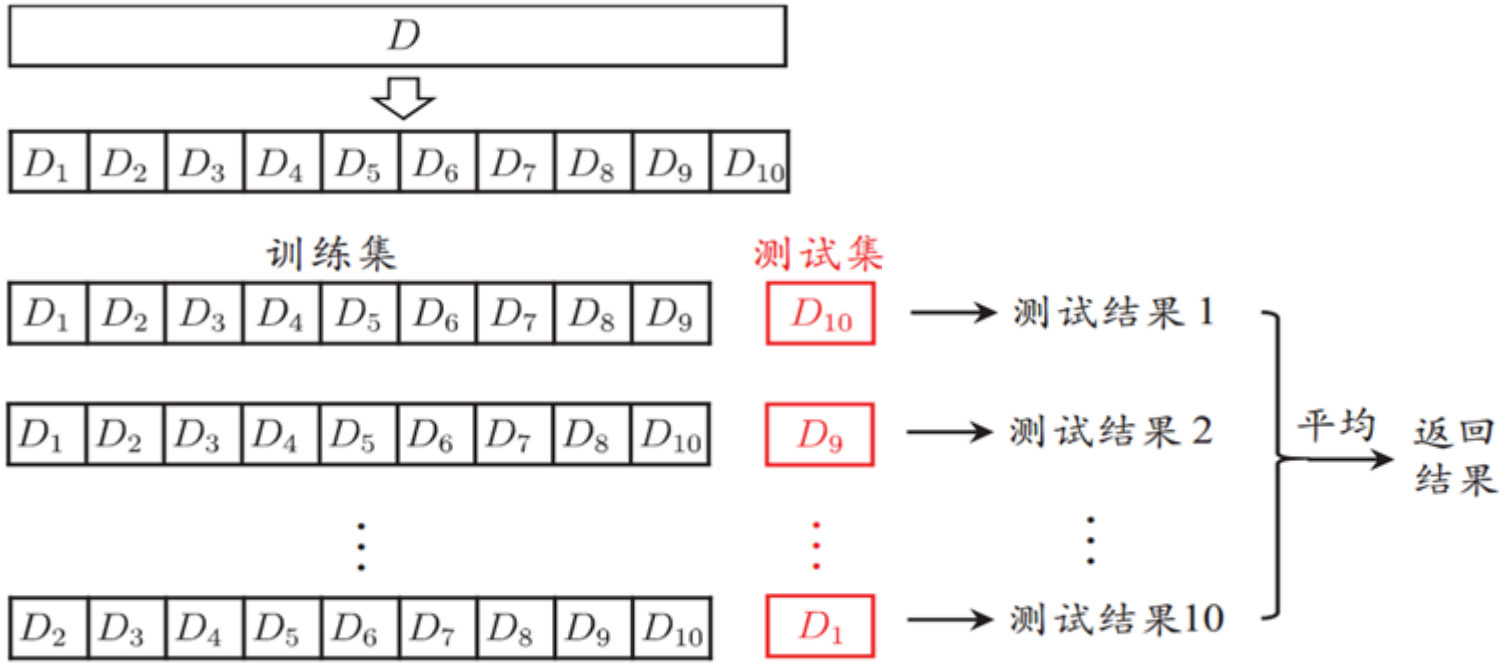
划分数据集，测试集数据占全集的30%：

```
train, test = train_test_split(data, test_size=0.3)
```

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html

超越单个测试集：交叉验证

将数据集分层采样划分为k个大小相似的互斥子集，每次用k-1个子集的并集作为训练集，余下的子集作为测试集，最终返回k个测试结果的均值，k最常用的取值是10.



10 折交叉验证示意图

与留出法类似，将数据集D划分为k个子集同样存在多种划分方式，为了减小因样本划分不同而引入的差别，k折交叉验证通常随机使用不同的划分重复p次，最终的评估结果是这p次k折交叉验证结果的均值，例如常见的“10次10折交叉验证”

假设数据集D包含m个样本，若令 $k = m$ ，则得到**留一法**：

- 不受随机样本划分方式的影响
- 结果往往比较准确
- 当数据集比较大时，计算开销难以忍受

超越单个测试集：交叉验证

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6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
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训练数据

验证数据

超越单个测试集：交叉验证

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7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
9	2013-03-08	Oz The Great and Powerful	215000000	234911825	Sam Raimi	PG	127
10	2013-05-16	Star Trek Into Darkness	190000000	228778661	J.J. Abrams	PG-13	123
11	2013-11-08	Thor: The Dark World	170000000	206362140	Alan Taylor	PG-13	120
12	2013-06-21	World War Z	190000000	202359711	Marc Forster	PG-13	116
13	2013-03-22	The Croods	135000000	187168425	Kirk De MiccoChris Sanders	PG	98
14	2013-06-28	The Heat	43000000	159582188	Paul Feig	R	117
15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

训练数据1

验证数据1

超越单个测试集：交叉验证

	Date	Title	Budget	DomesticTotalGross	Director	Rating	Runtime
0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
1	2013-05-03	Iron Man 3	200000000	409013994	Shane Black	PG-13	129
2	2013-11-22	Frozen	150000000	400738009	Chris BuckJennifer Lee	PG	108
3	2013-07-03	Despicable Me 2	76000000	368061265	Pierre CoffinChris Renaud	PG	98
4	2013-06-14	Man of Steel	225000000	291045518	Zack Snyder	PG-13	143
5	2013-10-04	Gravity	100000000	274092705	Alfonso Cuaron	PG-13	91
6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
9	2013-03-08	Oz The Great and Powerful	215000000	234911825	Sam Raimi	PG	127
10	2013-05-16	Star Trek Into Darkness	190000000	228778661	J.J. Abrams	PG-13	123
11	2013-11-08	Thor: The Dark World	170000000	206362140	Alan Taylor	PG-13	120
12	2013-06-21	World War Z	190000000	202359711	Marc Forster	PG-13	116
13	2013-03-22	The Croods	135000000	187168425	Kirk De MiccoChris Sanders	PG	98
14	2013-06-28	The Heat	43000000	159582188	Paul Feig	R	117
15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

训练数据2

验证数据2

超越单个测试集：交叉验证

	Date	Title	Budget	DomesticTotalGross	Director	Rating	Runtime
0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
1	2013-05-03	Iron Man 3	200000000	409013994	Shane Black	PG-13	129
2	2013-11-22	Frozen	150000000	400738009	Chris BuckJennifer Lee	PG	108
3	2013-07-03	Despicable Me 2	76000000	368061265	Pierre CoffinChris Renaud	PG	98
4	2013-06-14	Man of Steel	225000000	291045518	Zack Snyder	PG-13	143
5	2013-10-04	Gravity	100000000	274092705	Alfonso Cuaron	PG-13	91
6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
9	2013-03-08	Oz The Great and Powerful	215000000	234911825	Sam Raimi	PG	127
10	2013-05-16	Star Trek Into Darkness	190000000	228778661	J.J. Abrams	PG-13	123
11	2013-11-08	Thor: The Dark World	170000000	206362140	Alan Taylor	PG-13	120
12	2013-06-21	World War Z	190000000	202359711	Marc Forster	PG-13	116
13	2013-03-22	The Croods	135000000	187168425	Kirk De MiccoChris Sanders	PG	98
14	2013-06-28	The Heat	43000000	159582188	Paul Feig	R	117
15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

验证数据3

训练数据3

超越单个测试集：交叉验证

	Date	Title	Budget	DomesticTotalGross	Director	Rating	Runtime
0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
1	2013-05-03	Iron Man 3	200000000	409013994	Shane Black	PG-13	129
2	2013-11-22	Frozen	150000000	400738009	Chris BuckJennifer Lee	PG	108
3	2013-07-03	Despicable Me 2	76000000	368061265	Pierre CoffinChris Renaud	PG	98
4	2013-06-14	Man of Steel	225000000	291045518	Zack Snyder	PG-13	143
5	2013-10-04	Gravity	100000000	274092705	Alfonso Cuaron	PG-13	91
6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
9	2013-03-08	Oz The Great and Powerful	215000000	234911825	Sam Raimi	PG	127
10	2013-05-16	Star Trek Into Darkness	190000000	228778661	J.J. Abrams	PG-13	123
11	2013-11-08	Thor: The Dark World	170000000	206362140	Alan Taylor	PG-13	120
12	2013-06-21	World War Z	190000000	202359711	Marc Forster	PG-13	116
13	2013-03-22	The Croods	135000000	187168425	Kirk De MiccoChris Sanders	PG	98
14	2013-06-28	The Heat	43000000	159582188	Paul Feig	R	117
15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

验证数据4

训练数据4

交叉验证的语法

导入交叉验证函数：

```
from sklearn.model_selection import cross_val_score
```

用一个给定的模型执行交叉验证：

```
cross_val = cross_val_score(KNN, X_data, y_data, cv=4,
                            scoring='neg_mean_squared_error')
```

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.cross_val_score.html

交叉验证的语法

导入划分训练集和测试集的函数：

```
from sklearn.model_selection import cross_val_score
```

用一个给定的模型执行交叉验证：

```
cross_val = cross_val_score(KNN, X_data, y_data, cv=4,
                            scoring='neg_mean_squared_error')
```

cv的可能取值：

- None, to use the default 3-fold cross validation,
- integer, to specify the number of folds in a `(Stratified)KFold`,
- CV splitter,
- An iterable yielding (train, test) splits as arrays of indices.

其他CV splitter:

`LeaveOneOut, ShuffleSplit, StratifiedShuffleSplit` ……

http://scikit-learn.org/stable/modules/cross_validation.html

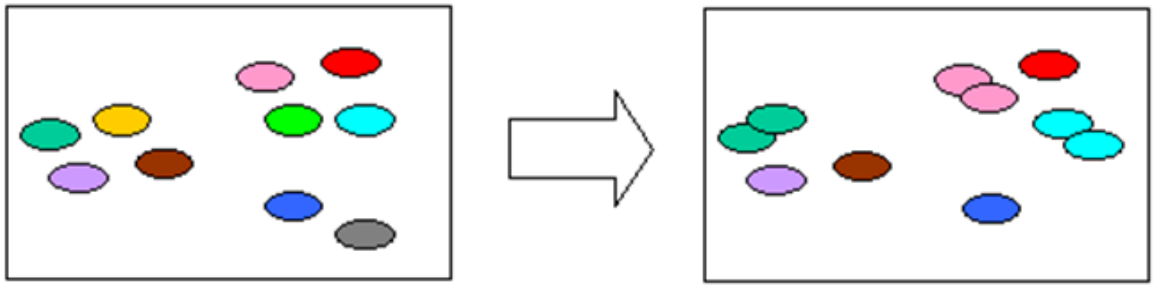
自助法：

以自助采样法为基础，对数据集 D 有放回采样 m 次得到训练集 D'，$D \setminus D'$ 用做测试集。

$$\lim_{m \mapsto \infty} \left(1 - \frac{1}{m}\right)^m \mapsto \frac{1}{e} \approx 0.368$$

- 实际模型与预期模型都使用 m 个训练样本
- 约有1/3的样本没在训练集中出现

- 从初始数据集中产生多个不同的训练集，对集成学习有很大的好处
- 自助法在数据集较小、难以有效划分训练/测试集时很有用；
- 由于改变了数据集分布可能引入估计偏差，在数据量足够时，留出法和交叉验证法更常用。



模型选择

多种学习算法

模型（同一算法下的不同参数配置）

“调参”与最终模型

模型的参数：一般由学习确定

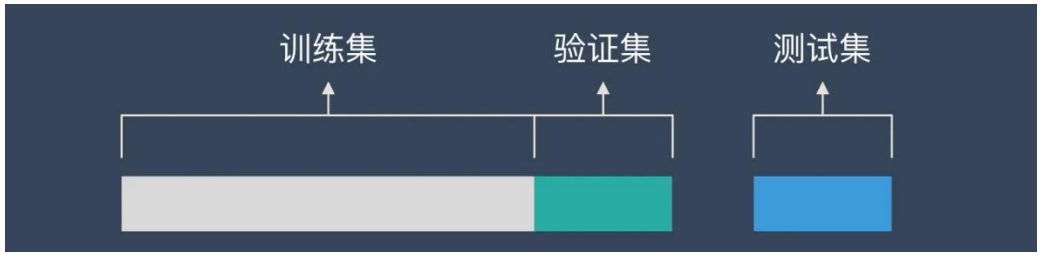
算法的参数：一般由人工设定，亦称“超参数”

调参过程：先产生若干模型，然后基于某种评估方法进行选择

参数调得好不好往往对最终性能有关键影响

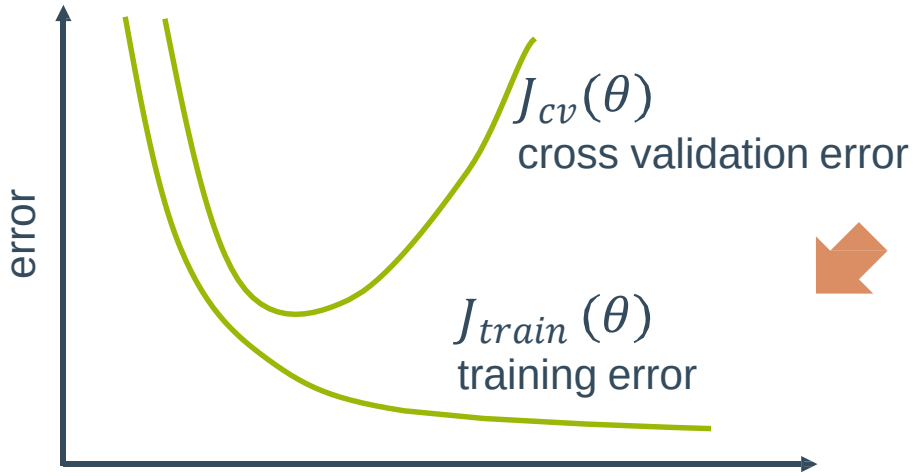
通常将包含个m样本的数据集$D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$
拆分成**训练集**S,**验证集**V和**测试集**T：

验证集（validation set）:模型评估与选择中用于评估的数据集常称为“验证集（validation set）”

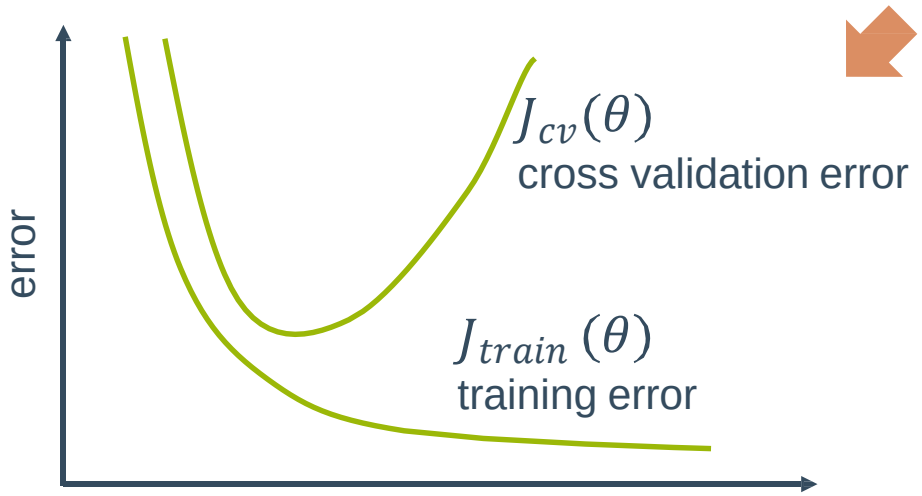


算法参数选定后，要用“训练集+验证集”重新训练最终模型

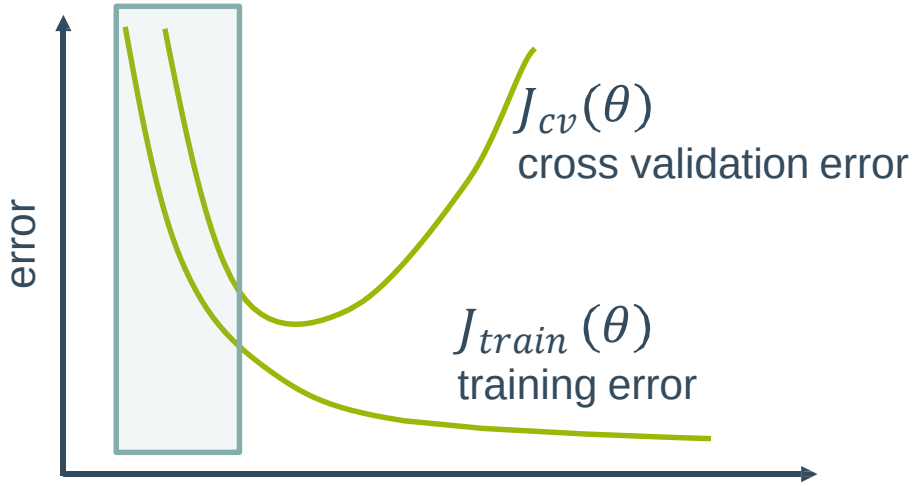
模型复杂度与误差

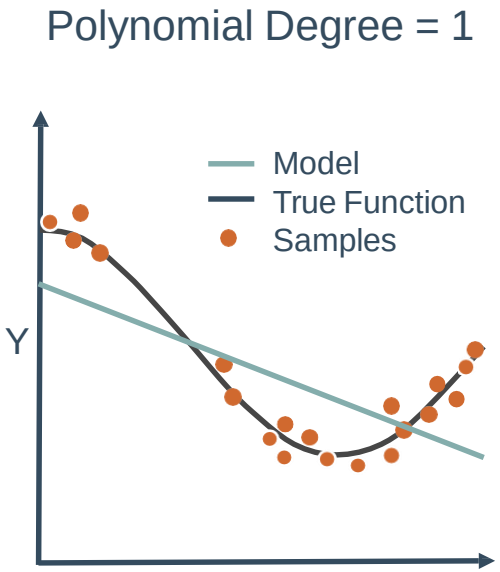

error
$J_{cv}(\theta)$
cross validation error
$J_{train}(\theta)$
training error

模型复杂度与误差


error
$J_{cv}(\theta)$
cross validation error
$J_{train}(\theta)$
training error

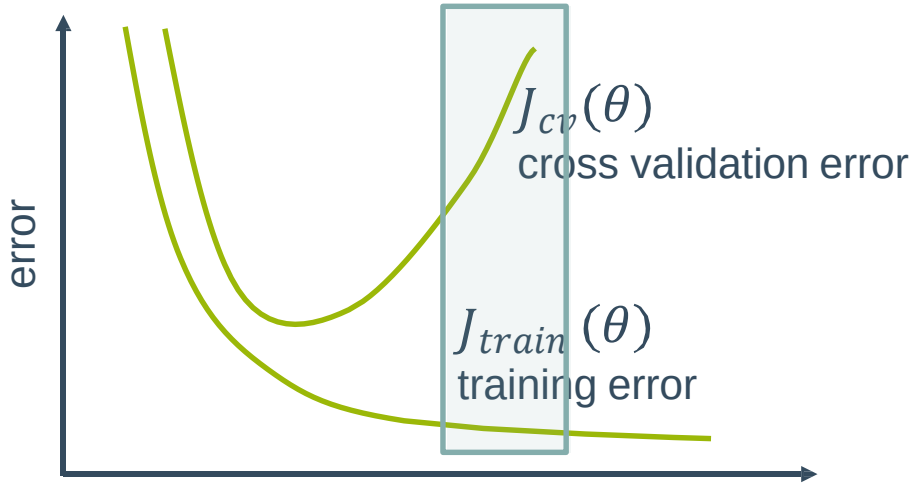
模型复杂度与误差

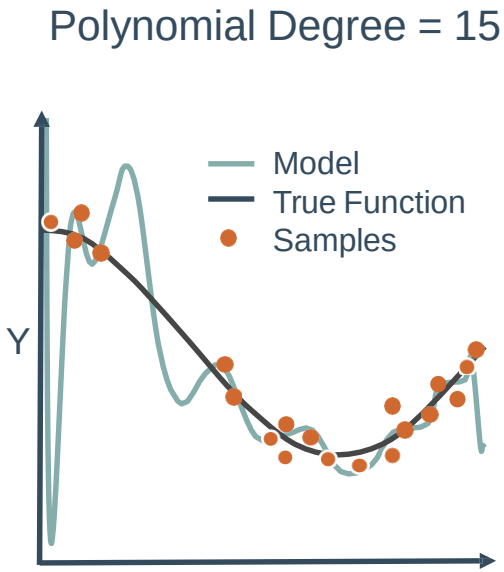




欠拟合：训练误差和交叉验证误差都很高

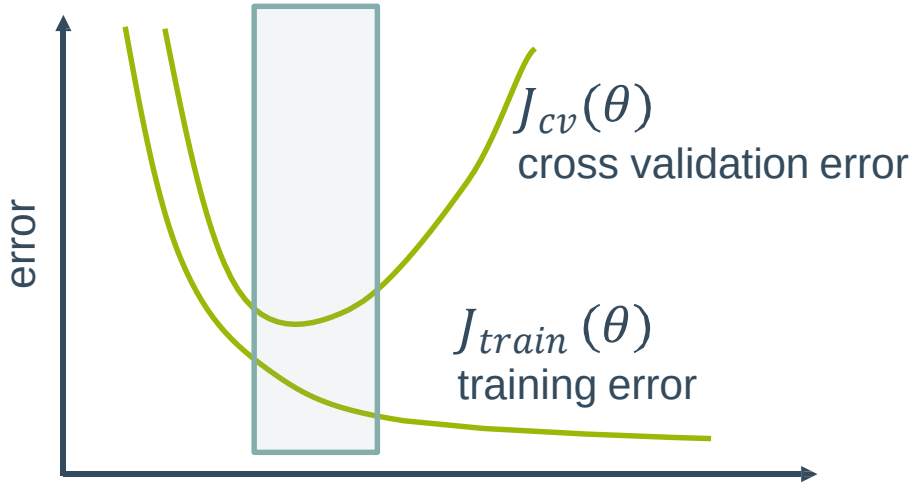
模型复杂度与误差

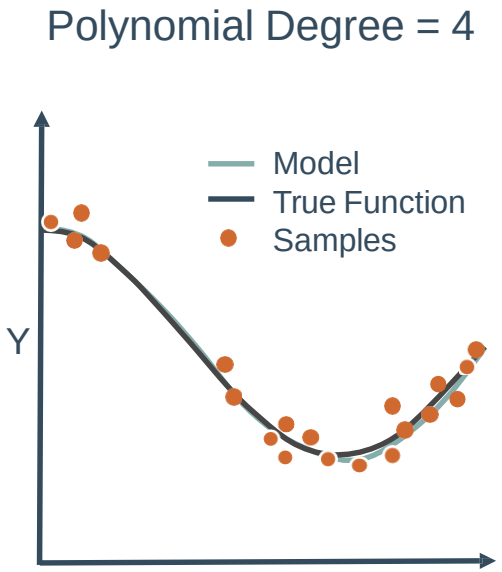




过拟合：训练误差低，交叉验证误差高

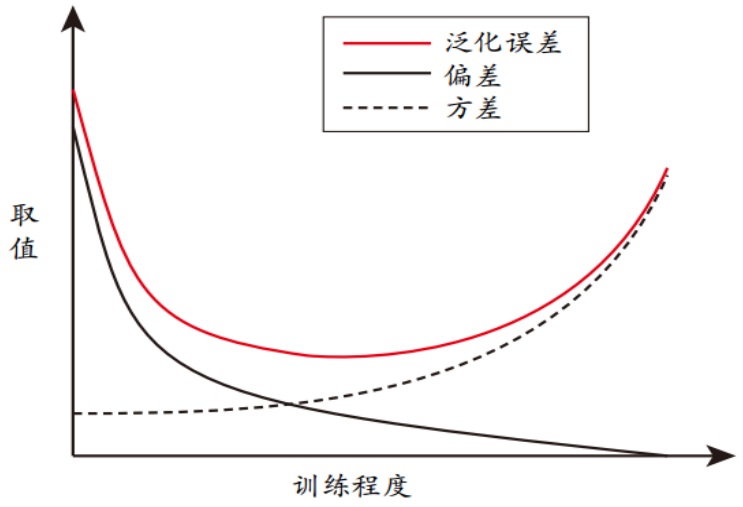
模型复杂度与误差


error
$J_{cv}(\theta)$
cross validation error
$J_{train}(\theta)$
training error


Polynomial Degree = 4
Model
True Function
Samples
Y

如图所示，假如我们能控制算法的训练程度：

- 在训练不足时，学习器拟合能力不强，训练数据的扰动不足以使学习器的拟合能力产生显著变化，此时偏差主导泛化错误率；
- 随着训练程度加深，学习器拟合能力逐渐增强，方差逐渐主导泛化错误率；
- 训练充足后，学习器的拟合能力非常强，训练数据的轻微扰动都会导致学习器的显著变化，若训练数据自身非全局特性被学到则会发生过拟合。



泛化误差与偏差、方差的关系示意图

模型选择

三个关键问题:

- 如何获得测试结果? ⇨ 评估方法
- 如何评估性能优劣? ⇨ 性能度量

性能度量

性能度量是衡量模型泛化能力的评价标准，反映了任务需求；

使用不同的性能度量往往会导致不同的评判结果

什么样的模型是“好”的，不仅取决于算法和数据，还取决于任务需求

在预测任务中，给定样例集 $D=\{(\boldsymbol{x}_1,y_1),(\boldsymbol{x}_2,y_2),\ldots,(\boldsymbol{x}_m,y_m)\}$
评估学习器 f 的性能也即把预测结果 $f(\boldsymbol{x})$ 和真实标记比较.

回归任务最常用的性能度量是“均方误差”：

$$E(f;D)=\frac{1}{m}\sum_{i=1}^{m}\left(f\left(\boldsymbol{x}_i\right)-y_i\right)^2$$

模型选择

三个关键问题:

- 如何获得测试结果? ⇨ 评估方法
- 如何评估性能优劣? ⇨ 性能度量
- 如何判断实质差别? ⇨ 比较检验

在某种度量下取得评估结果后，是否可以直接比较以评判优劣？

- 测试性能并不等于泛化性能
- 测试性能随着测试集的变化而变化
- 很多机器学习算法本身有一定的随机性

直接选取相应评估方法在相应度量下比大小的方法不可取！

机器学习 ⇒ **“概率近似正确”**

$$P(|f(\boldsymbol{x}) - y| \leq \epsilon) \geq 1 - \delta$$

假设检验为学习器性能比较提供了重要依据，基于其结果我们可以推断出若在测试集上观察到学习器A比B好，则A的泛化性能是否在统计意义上优于B，以及这个结论的把握有多大。

常用方法

统计假设检验 (hypothesis test) 为学习器性能比较提供了重要依据

- 两学习器比较
  - 交叉验证 t 检验 (基于成对 t 检验)

    k 折交叉验证； 5x2交叉验证
  - McNemar 检验 (基于列联表，卡方检验)
- 多学习器比较
  - Friedman + Nemenyi
    - Friedman检验 (基于序值，F检验; 判断”是否都相同”)
    - Nemenyi 后续检验 (基于序值，进一步判断两两差别)

统计显著性